

The accessibility engineer that fixes your code. And proves it.

An agentic workflow that repairs WCAG violations in real website source — then verifies every fix with deterministic re-scans, content-integrity checks and localized visual diffs, rolls back anything destructive, and routes judgment calls to a human with drafted fixes. The output: a fixed site and evidence a business owner can hand to their lawyer.

134 / 134

seeded WCAG violations
remediated & independently
verified

0

hallucinated image descriptions
(the baseline shipped 8 across
two runs)

32

scanner-invisible issues found &
drafted for human review

~\$0.50

measured compute per site —
vs \$350–550/page manual
remediation

A lawsuit, not a lint warning

The typical defendant in a web-accessibility lawsuit is not a tech company. It's a bakery, a dental clinic, a small law firm — a site built by a freelancer in 2019 that nobody has touched since. The owner discovers the problem when the demand letter arrives.

5,114

US web accessibility lawsuits in 2025 — a record (UsableNet year-end report)

64%

of 2025 defendants had revenue under \$25M — small businesses

95.9%

of the top 1M homepages fail WCAG; 56 errors per page on average (WebAIM 2026)

Jun 2025

European Accessibility Act in force — fines to €100k (DE) or 5% of turnover (IT)

Behind the legal risk sits the actual point: ~87 million people in the EU alone live with a disability. For them a missing form label isn't a compliance item — it's a door that doesn't open.

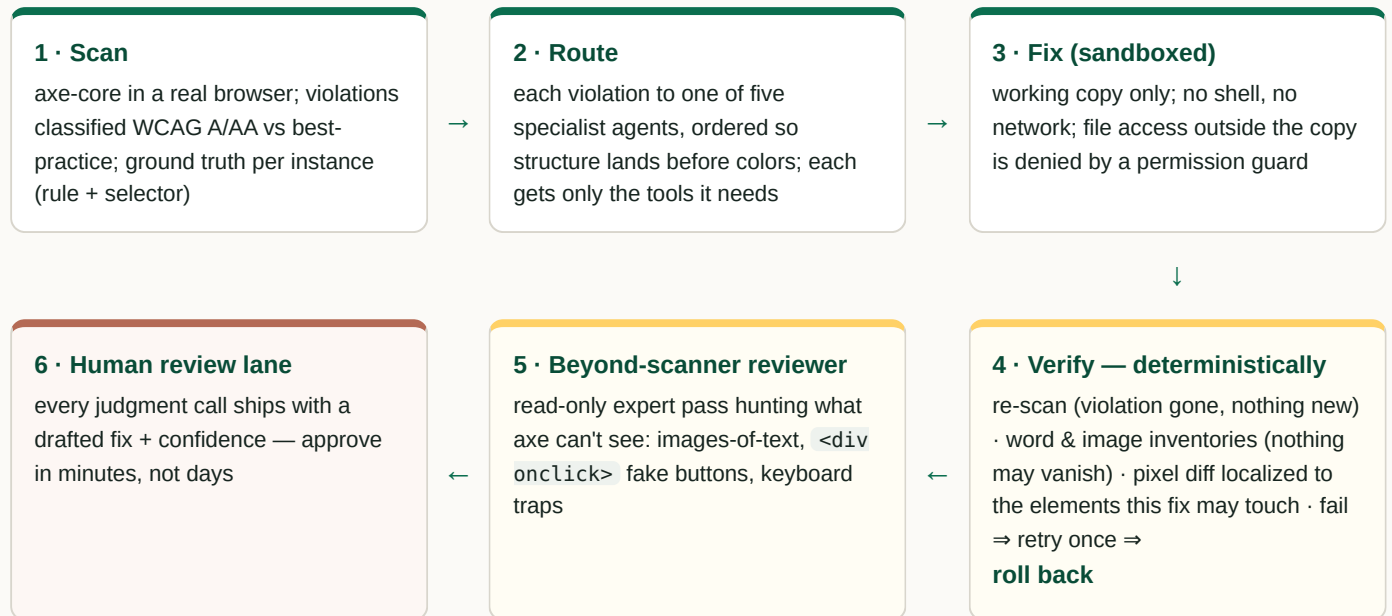
Why today's options fail this user

	WHAT IT DOES	WHY IT FAILS
Overlay widgets	masks issues in a JS layer — code untouched	FTC fined accessiBe \$1M (2025) over compliance claims; ~25% of sued companies had a widget installed <i>when sued</i> ; 1,031 practitioners signed the Overlay Fact Sheet against them
Scanners	finds issues, fixes nothing	the report is homework; automation detects ~57% of issues by volume (Deque) and none of the fixes
Consultants	fixes properly	\$1,250–2,750 audit + \$350–550 <i>per page</i> remediation, weeks of lead time — priced for enterprises
Paste into a chatbot	sometimes brilliant, sometimes destructive	we measured it: same clinic page, two runs — one hid the clinic's booking phone number from blind users, one didn't. Nothing tells you which run you got.

The bottleneck in 2026 is not finding problems — it's not even fixing most of them. It's knowing, with evidence, that they were fixed and nothing was silently broken — at a price a bakery can pay.

Agents with a deterministic spine

Language models supply judgment. Everything checkable is checked by machinery the model cannot argue with — and every design choice answers a *documented* failure mode of LLM accessibility repair (research mapping in `docs/research/prior-art.md`).



The three principles

- **Models judge; tools know.** Contrast fixes come from WCAG luminance math that finds the closest compliant color *preserving the brand hue*. Alt text may only be written after the agent has rendered and **looked at** the image. Perception precedes assertion — so the hallucination class is eliminated, not just caught.
- **The verifier is not the fixer.** Acceptance is decided by axe re-scans and content inventories; failed fixes roll back automatically and join the human queue.
- **Honesty is the product.** Automation covers ~57% of issues by volume. StepFree claims verified fixes for the detectable layer and routes the rest to humans — the claim discipline the FTC fined the market leader for lacking.

Memory, skills, orchestration — used purposefully

- **Memory:** a conventions ledger — the first agent to resolve a color pair or set an alt style records it; every later fixer on every page reuses it. One accessible palette per site, not one per page.
- **Skills:** five per-domain knowledge files encode WCAG craft (e.g. "no ARIA beats bad ARIA — prefer removing it"; "an image inside a link describes the destination").
- **Orchestration:** a deterministic pipeline; every changelog stage is a replayable flag (`--stage v1...final`), and every agent run is captured as a full trajectory.

The finding we didn't expect — reported plainly

Corpus: 12 synthetic small-business sites (13 pages), **134 scanner-verified WCAG A/AA violations** mirroring the WebAIM distribution, ground truth locked before any run. Baseline = the same model (`claude-sonnet-5`), one direct prompt. The evaluator is independent of the system under test: it re-scans and re-screenshots every page itself.

On small static pages, the one-shot baseline remediated **134/134** scanner-detectable instances with zero regressions — frontier-model capability on the *scanner-visible* layer is saturated, and we say so. The layers that decide whether a real person can use the site are where the systems separate:

	BASELINE — ONE DIRECT PROMPT	STEPFREE — FINAL
WCAG A/AA instances remediated & verified	134/134	134/134 · 13/13 pages shippable
Alt-text truthfulness (0–2, vision-judged by Claude Opus)	1.53 / 1.37 across two runs — 4 hallucinated or destructive alts each run	1.80 (v3: 1.83) — zero hallucinations
Informative content silenced with <code>alt=""</code>	1 of 2 runs — including the clinic's booking phone number	0 — the image agent must look before writing
Scanner-invisible issues surfaced for humans	0 (by definition)	32, each with a drafted fix
Evidence produced	none — ship on faith	audit report + per-fix verification + full trajectories
Measured compute, whole corpus	\$2.03	\$5.78 (~\$0.48/site)

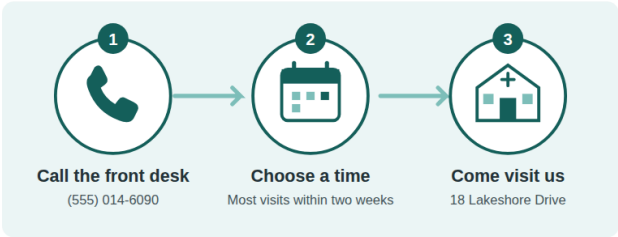


Exhibit A. The baseline resolved this image's missing-alt violation by marking it decorative (`alt=""`) — hiding the phone number and address from screen-reader users. The scanner scored the page as *improved* (11 → 0). StepFree's vision-grounded agent read the image and wrote alt text carrying the steps; run two of the baseline described it but dropped the phone number. Same input, coin-flip outcomes — unless something checks.

Other baseline exhibits (judge's reasoning on file)

- "Two wine glasses beside an uncorked bottle" — image shows **one** glass and a **sealed** bottle.
- "Officials break ground with ceremonial shovels..." — an entirely invented ceremony; the image is a building.
- An image-of-text hero given alt that doesn't match the words in the image (WCAG 1.4.5).

Economics per 13-page site

	HUMAN TIME	COST
Manual remediation	~28 h	\$4,550–7,150 (at \$350–550/page)
Baseline + due-diligence QA	~4.3 h	\$2 + QA labor
StepFree	~1 h (review queue + report)	\$5.78 (~\$0.48/site) compute

Assumptions stated in docs/research/cost-model.md; primary metric unaffected by them.

Six measured stages, one removed experiment

STAGE	WHAT CHANGED & WHY	EVIDENCE (SAME EVAL, ALL 134 INSTANCES)	DECISION
baseline	one direct prompt — what a busy developer does today	134/134 on the scanner layer, but alt truthfulness 1.53/1.37 with 4 hallucinations per run; booking infographic silenced in run 1	the problem moved from capability to groundedness & trust
v1	basic agent: axe report + file tools	134/134 · \$1.48 — cheaper than baseline: targeted edits beat full-page rewrites	kept as chassis
v2	deterministic verify loop (re-scan diff, bounded retry)	134/134; retries fired 0 times on-corpus	kept — converts "probably fine" into "proven"
v3	5 specialists + knowledge files + contrast math + mandatory image viewing	alt truthfulness 1.83, zero hallucinations ; silenced infographic 0 → 2	kept — the load-bearing change
v4	integrity guardrails: word/image inventories, localized pixel diff, auto-rollback	134/134, 0 rollbacks needed — the reason "0 pages damaged" is a fact, not an assumption	kept — quiet gates still testify
final	conventions memory + beyond-scanner review lane + evidence report	134/134 · 32 review items incl. all 7 planted scanner-invisible traps (7/7 recall) · \$5.78 corpus	the product

The experiment we removed: a strict visual-diff gate

"Reject any fix changing >0.5% of pixels" sounds safe. A controlled, LLM-free experiment (`stepfree/eval/pixel-vs-harm.mjs`) shows it's broken both ways:

EDIT	VIEWPORT PIXELS CHANGED	WORDS LOST
Contrast recolor (legitimate)	0.27%	0
Entire menu item deleted (destructive)	1.32%	13
Naive heading retag (legitimate)	1.93%	0

Pixel area measures reflow, not harm — any gate loose enough to admit the legitimate retag admits the deletion. Removed as a gate, kept as a localized advisory; the word-inventory check orders harm correctly.

Hot take. We built verification armor expecting to catch a clumsy model — and the model never stumbled where we watched. It failed by *asserting what it had no way to know*: describing images it never saw, silencing an infographic with a phone number in it. In 2026, agentic scaffolding isn't primarily there to make models capable — it's there to make failures visible, bounded, and impossible to ship silently. Even our tooling agreed: shown an unviewable image, our grounded judge said *"I cannot honestly score this"* — the ungrounded baseline invented a croissant.

Groundedness is an architecture property, not a model property. Capability saturated; trust didn't. Sell trust.

Run it yourself in an afternoon

Clean-environment reproduction

```
git clone github.com/shil720/micro1-hackathon
cd stepfree && npm install && npx playwright
install chromium
export ANTHROPIC_API_KEY=...
node stepfree/eval/selftest.mjs #
deterministic core
node stepfree/eval/run-eval.mjs verify-
fixtures # lock ground truth
node stepfree/src/cli.mjs fix fixtures/08-
shelter --out runs/demo
bash stepfree/eval/run-all-stages.sh 3 # full
evaluation
```

- Node 22 · Playwright 1.62 · axe-core (via @axe-core/playwright 4.13) · Claude Agent SDK
- Full ladder ≈ 2.5–4 h, ≈ \$18 of API compute; single-site demo ≈ 6 min, ≈ \$1
- All data synthetic & in-repo; scanner + integrity checks fully deterministic; the evaluator refuses to run if fixture ground truth drifts

Deliverables in the repository

- **README.md** — problem, user, solution, results
- **CHANGELOG.md** — every iteration with evidence
- **EVALUATION.md** — metrics defined up front, full tables, limitations
- **REPRODUCTION.md** — this page, expanded
- **VIDEO_SCRIPT.md** — the 5-minute video, verbatim
- **trajectories/** — every agent type, readable + raw JSONL
- **reports/** — a generated audit report (it scans 0/0 itself)

Why this is a company, not a demo

- **Wedge:** \$149 one-time "fix & prove" per site · \$49/mo monitoring · \$499/mo agency tier (20 sites). Measured COGS ≈ \$0.50/site → >90% gross margin.
- **First buyers:** web agencies — thousands of 2019-era SMB sites, already fielding the demand-letter call.
- **Market:** accessibility software → ~\$1B by 2030; incumbents at \$40–50M ARR prove willingness to pay — for products that don't fix code. FTC action + 1,000+ suits/yr against widget users are the tailwind for "actually fix it".
- **Moat:** anyone can prompt a model to edit HTML. The verification stack, evidence trail, and honesty posture are what a stranger can trust with production — and what survives a legal letter.

Safety & ground rules

- All edits in a sandboxed working copy; deploying is a human act. Agents: no shell, no network, path-guarded file access.
- Human reviewer in the loop for every judgment call, with drafted fixes.
- Synthetic data only; no real business named; every claim traces to a source or a JSON artifact in the repo.
- Built openly with the tools it evaluates: Claude agents did the heavy lifting under Shivam's direction; every number comes from the deterministic harness.

Curb cuts were built for wheelchairs — everyone with a stroller or a suitcase uses them. Fix the code, and everybody walks in. **This is StepFree.**